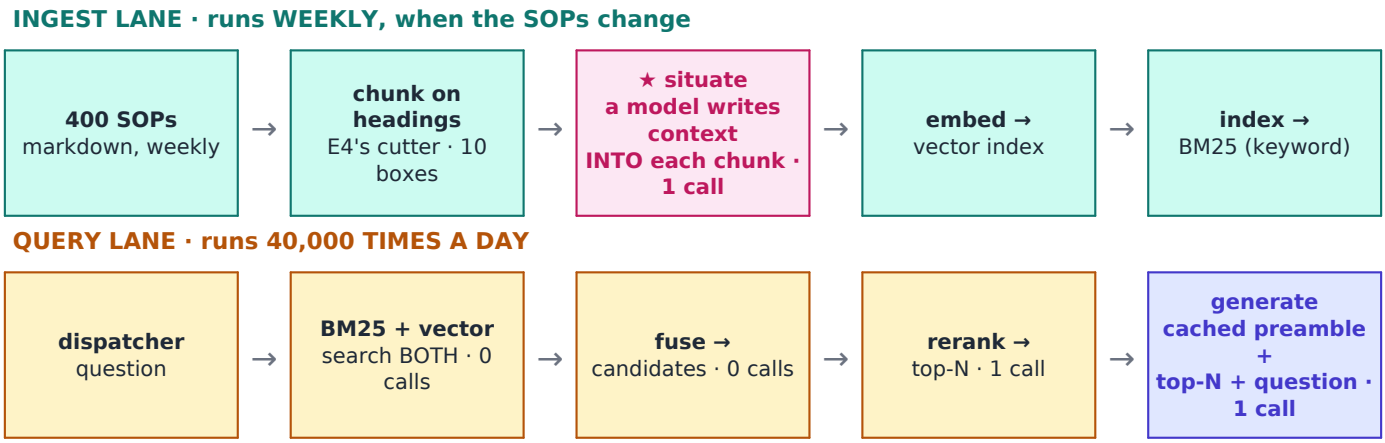
E10 · Two Lanes, Explained

The Part 4 drawing of the Loading Dock, run as code against SOP 214 on Claude Fable 5.1, and read back in five parts. Five model calls in all: one in the weekly lane, two per question in the daily lane.



The star is contextual retrieval: context is written into each chunk BEFORE both indexes see it. The two lanes run on two different clocks, and that is the whole lesson.

PART 1 The ingest lane

runs once, before any question

```
=== INGEST LANE (runs when the SOPs change) ===
chunked on headings : 10 chunks
situated            : 10 context sentences written, one per chunk (1 model call)
indexed             : vectors x10, BM25 vocabulary 374 words (plain) -> 416 words (situated)

The sentence situate wrote in front of section 4:
  SOP 214 section 4 gives the six-step procedure for re-filing or extending an e-way bill after a hold,
  i.e. fixing the paperwork so a stuck shipment can move again.
```

This is the top row of the drawing, run once. The SOP was cut into ten boxes on its headings, the same cut E4 made. Then **one Fable call** read all ten and wrote one sentence of context for each. Then the two indexes were built from the chunks **with those sentences glued on the front**. No question was asked. Everything here is paid for before any dispatcher types anything.

THE TELL 374 words → 416 words. Forty-two new words entered the register's index, and two of them are **paperwork** and **stuck**. SOP 214 never says either word. Fable was told to use the words a customer would use, and it did. Those two words decide the second question.

PART 2 Query lane · first question · retrieval

the SOP's own words

```
=== QUERY LANE · 'how do I re-file an e-way bill?' ===
where section 4 (re-filing) ranks, plain shelves -> situated shelves:
  BM25   : #1 -> #1
  vector : #2 -> #2
fused candidates    : §4. Re-filing..., §3. Cancellation..., §5. Vehicle..., §2. First checks..., §Preamble ...
reranked top-3      : §4. Re-filing..., §2. First checks..., §3. Cancellation...
```

The question uses the SOP's words, so the register wins outright and situating changes nothing. That is correct: it is insurance for hard questions, not a boost for easy ones. Fusion put §3 (cancellation) second because §3 and §4 share most of their vocabulary, the E4 seam again. The **reranker read the text** of each candidate against the question and demoted cancellation to third. A reranker re-sorts the shortlist by reading, not by word counts.

PART 3 The first answer and its bill

generate · the cache is written

```
{"stage": "generate", "input_tokens": 1338, "output_tokens": 398, "cache_write": 1439, "cache_read": 0}
answer:
  Re-filing applies when goods must still move but the bill is expired ... [SOP 214 - 4]. Steps:
  1. Pull the officer's remark ... [SOP 214 - 4].
  ...
  6. Email the new bill number to the depot officer ... the hold lifts only on officer confirmation [SOP 214 - 4
].
  Before starting, complete section 2 checks ... [SOP 214 - 2]. Don't cancel during a hold [SOP 214 - 3].
```

All six steps, every fact with a chunk label in brackets as the rules demanded, and the two supporting chunks used sensibly: do the checks first, do not cancel. Compare E4's fixed-box answer with four steps MISSING. Same model, same SOP. The difference is that shelving and retrieval handed it the whole section.

THE BILL **cache_write 1,439**: the preamble (desk rules + a catalogue of all ten chunks) was written into the cache on this first question. **cache_read 0**: nothing was there yet. The 1,338 input tokens are the part that changes every time, the three chunks and the question. Hold these numbers for Part 4.

PART 4 Second question · where the star earns its name

the customer's words

```
=== QUERY LANE · 'parcel stuck, paperwork problem, what should I do' ===
where section 4 (re-filing) ranks, plain shelves -> situated shelves:
  BM25   : #6 -> #3
  vector : #6 -> #4
fused candidates    : §1. What a customs hold is, §Preamble, §4. Re-filing..., §8. Message templates, §7. Escala
tion ...
reranked top-3      : §2. First checks..., §4. Re-filing..., §1. What a customs hold is
{"stage": "generate", "input_tokens": 1241, "output_tokens": 375, "cache_write": 0, "cache_read": 1439}
```

None of parcel, stuck or paperwork appears anywhere in SOP 214. On the plain shelves §4 sat **sixth on both indexes**: a plain system would never have shown it to the model at top-3. On the situated shelves it rose to third and fourth, entirely because of the sentence from Part 1. Three stages then did three different jobs: **situating got §4 into the race, fusion kept it in the shortlist, reranking placed it**, second behind 'first checks', which is the right order for someone who only knows the parcel is stuck.

Part 4, in numbers

	plain shelves	situated shelves	what moved it
Q1 'how do I re-file an e-way bill?' · BM25	#1	#1	Nothing. The SOP's own words: the register wins outright.
Q1 · vector	#2	#2	Nothing. Situating is insurance for hard questions, not a boost for easy ones.
Q2 'parcel stuck, paperwork problem' · BM25	#6	**#3**	'stuck' and 'paperwork' now sit in §4's text: the register matches them.
Q2 · vector	#6	**#4**	The toy vector maps stuck → hold and finds the sentence's 'hold'.

The cache story, closed

generate call	input_tokens (changes)	cache_write	cache_read	meaning
Question 1	1,338	**1,439**	0	The preamble (rules + catalogue of all 10 chunks) is written into the cache. Nothing to read yet.
Question 2	1,241	0	**1,439**	The whole preamble comes back from the cache at a tenth of the input price. Only the chunks and the question are paid in full.

PART 5 **The two clocks, and the answer that went with them** why situate is a bargain

```
answer:
  Start by confirming the consignment id ... read the hold code (EWB = e-way bill, INV = invoice, VEH = vehicle)
 ... [SOP 214 - 1].
  Compare the e-way bill number against the invoice ... [SOP 214 - 2].
  Then choose the route: ... bill expired or invoice value wrong → section 4 re-filing ... [SOP 214 - 2].
  If re-filing, follow the six steps ... do not tell the customer it's cleared until the officer confirms [SOP 2
14 - 4].

=== THE TWO CLOCKS ===
ingest lane : 1 situate call per ingest run. 400 SOPs, weekly: a few hundred calls a week, paid once.
query lane  : 2 calls per question (rerank, generate). 40,000 questions a day: 80,000 calls a day.
```

The second answer is a triage, not a procedure, and that is the right shape for a vague question: identify the hold code, run the checks, pick the route, then re-file. It cites three chunks and ends with the one promise the SOP forbids making early.

The two clocks are the whole point of the drawing. Anything moved from the bottom clock to the top clock is nearly free, and situating is the biggest such move: one sentence written at ingest improves every one of the forty thousand searches that follow. The cache is the same idea inside the query lane: the stable part of the prompt is paid once and read cheaply thereafter.

5
Fable calls in the whole run: 1 situate, 2 rerank, 2 generate

#6 → #3
where §4 ranked for the customer's words, plain → situated (BM25)

1,439
preamble tokens written once, read back on question 2 at a tenth of the price

80,000
query-lane calls a day at 40,000 questions, against a few hundred ingest calls a week

FOR THE EXAM

Situate before **both** indexes, paid once at ingest. Fuse the two ranked lists. Retrieve wide, rerank narrow (150 → 20 in the book, 10 → 3 here). Cache the preamble. And the numbers from your own run: 35 → 49 → 67 percent failure cut in the book; §4 from sixth to third, and 1,439 tokens back from the cache, on your laptop.